

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Binomial Expansion

*Binomial questions are coefficient questions: choose the term,
then control powers.*

BINOMIAL EXPANSION · P2 · 04

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 TERM: General Term

TRIGGER *expand $(a + bx)^n$*

02 COEFF: Coefficient Match

TRIGGER *coefficient of x^k*

03 RANGE: Validity Range

TRIGGER *binomial approximation*

04 APPROX: Approximate A Number

TRIGGER *use binomial to estimate*

BANK EVIDENCE Local pages: 16 problem, 36 answer, 6 notes. Website primary entries: 19.

01 TERM: General Term

TRIGGER *expand* $(a + bx)^n$

BECOMES Use the binomial term formula.

FIRST LINE TO WRITE

$$T_{r+1} = \binom{n}{r} a^{n-r} (bx)^r$$

SIMPLEST STRATEGY

- 1 Take r from the term needed.
- 2 Enter binomial coefficient.
- 3 Raise both parts to powers.
- 4 Multiply and simplify.

WORKED MODEL

$$\text{In } (1 + 2x)^5, T_4 = \binom{5}{3} (1)^2 (2x)^3 = 80x^3.$$

HARD VARIANTS

- 1 For descending powers, solve the power of x first.
- 2 If the bracket contains $-x$, carry the sign with the term.
- 3 Use exact binomial coefficients, not calculator decimals.

— BOTTOM LINE

The term formula prevents missing coefficients.

03 RANGE: Validity Range

TRIGGER *binomial approximation*

BECOMES Require the small part to have magnitude less than 1.

FIRST LINE TO WRITE

$$|u| < 1$$

SIMPLEST STRATEGY

- 1 Rewrite as $(1 + u)^n$.
- 2 Ask what u is.
- 3 Note $|u| < 1$.
- 4 Give the interval.
- 5 Endpoints excluded.

WORKED MODEL

$$(1 - 3x)^{-2} \text{ is valid when } |-3x| < 1, \text{ so } |x| < \frac{1}{3}.$$

HARD VARIANTS

- 1 If the bracket is $a + bx$, factor out a first.
- 2 Strict inequality means endpoints are not included.
- 3 Use the range before trusting an approximation.

— BOTTOM LINE

The range belongs to the transformed small term.

04 APPROX: Approximate A Number

TRIGGER *use binomial to estimate*

BECOMES Choose x so the bracket matches the number.

FIRST LINE TO WRITE

$$(1 + u)^n \approx \text{first terms}$$

SIMPLEST STRATEGY

- 1 Align the number with the bracket.
- 2 Pick a small u .
- 3 Place u into expansion.
- 4 Round at the end.
- 5 Observe validity.
- 6 Xexclude over-precision.

WORKED MODEL

$$\sqrt{1.04} \approx 1 + \frac{1}{2}(0.04) - \frac{1}{8}(0.04)^2.$$

HARD VARIANTS

- 1 Factor constants so the remaining bracket is close to 1.
- 2 Use enough terms for the requested accuracy.
- 3 Check that the chosen u lies inside the validity range.

— BOTTOM LINE

Approximation begins by matching the bracket.

SAME IDEA Use APPROX: Approximate A Number.

QUESTION

Use binomial expansion to approximate $(1.02)^5$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
expand $(a + bx)^n$	$T_{r+1} = \binom{n}{r} a^{n-r} (bx)^r$
coefficient of x^k	coefficient of x^k
binomial approximation	$ u < 1$
use binomial to estimate	$(1 + u)^n \approx$ first terms